

High-Performance Order Execution and Management System

1.0

Generated by Doxygen 1.9.1

1 Class Index	1
1.1 Class List	1
2 File Index	3
2.1 File List	3
3 Class Documentation	5
3.1 DeribitClient Class Reference	5
3.1.1 Detailed Description	6
3.1.2 Constructor & Destructor Documentation	6
3.1.2.1 DeribitClient()	6
3.1.3 Member Function Documentation	7
3.1.3.1 cancel_all() [1/2]	7
3.1.3.2 cancel_all() [2/2]	7
3.1.3.3 cancel_order()	7
3.1.3.4 close()	8
3.1.3.5 connect()	8
3.1.3.6 generate_id()	8
3.1.3.7 get_orderbook()	8
3.1.3.8 get_positions()	9
3.1.3.9 handle_error()	9
3.1.3.10 init_logger()	9
3.1.3.11 list_subscriptions()	10
3.1.3.12 modify_order()	10
3.1.3.13 place_order()	10
3.1.3.14 send_auth()	11
3.1.3.15 start_server()	11
3.1.3.16 stop_server()	12
3.1.3.17 subscribe_book()	12
3.1.3.18 unsubscribe()	12
3.2 MessageParser Class Reference	12
3.2.1 Detailed Description	13
3.2.2 Constructor & Destructor Documentation	13
3.2.2.1 MessageParser()	13
3.2.3 Member Function Documentation	13
3.2.3.1 parse_and_print()	13
4 File Documentation	15
4.1 src/deribit_client.cpp File Reference	15
4.2 src/message_parser.cpp File Reference	15
4.2.1 Detailed Description	15

Chapter 1

Class Index

1.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

DeribitClient::ClientSession	??
DeribitClient	
Client class for interacting with the Deribit cryptocurrency exchange API	5
MessageParser	
Parser for handling Deribit WebSocket API messages Provides methods to parse and handle different types of responses from the exchange	12

Chapter 2

File Index

2.1 File List

Here is a list of all files with brief descriptions:

build_and_run.sh	??
main.cpp	??
build/CMakeFiles/3.22.1/CompilerIdC/ CMakeCCompilerId.c	??
build/CMakeFiles/3.22.1/CompilerIdCXX/ CMakeCXXCompilerId.cpp	??
build/CMakeFiles/3.31.4/CompilerIdC/ CMakeCCompilerId.c	??
build/CMakeFiles/3.31.4/CompilerIdCXX/ CMakeCXXCompilerId.cpp	??
build/CMakeFiles/DeribitClient.dir/ main.cpp.o.d	??
build/CMakeFiles/DeribitClient.dir/src/ deribit_client.cpp.o.d	??
build/CMakeFiles/DeribitClient.dir/src/ message_parser.cpp.o.d	??
include/ deribit_client.hpp	??
include/ message_parser.hpp	??
include/ prompts.hpp	??
include/ websocket_server.hpp	??
src/ deribit_client.cpp	
Implementation of the DeribitClient class for interacting with Deribit exchange	15
src/ message_parser.cpp	
Implementation of the MessageParser class for handling Deribit API messages	15

Chapter 3

Class Documentation

3.1 DeribitClient::ClientSession Struct Reference

Public Attributes

- `std::shared_ptr< websocket::stream< tcp::socket > >` [socket](#)
- `std::set< std::string >` [subscriptions](#)
- `std::chrono::steady_clock::time_point` [last_active](#)

3.1.1 Member Data Documentation

3.1.1.1 [last_active](#)

`std::chrono::steady_clock::time_point` `DeribitClient::ClientSession::last_active`

3.1.1.2 [socket](#)

`std::shared_ptr<websocket::stream<tcp::socket> >` `DeribitClient::ClientSession::socket`

3.1.1.3 [subscriptions](#)

`std::set<std::string>` `DeribitClient::ClientSession::subscriptions`

The documentation for this struct was generated from the following file:

- `include/deribit_client.hpp`

3.2 DeribitClient Class Reference

Client class for interacting with the Deribit cryptocurrency exchange API.

```
#include <deribit_client.hpp>
```

Collaboration diagram for DeribitClient:

3.3 MessageParser Class Reference

Parser for handling Deribit WebSocket API messages Provides methods to parse and handle different types of responses from the exchange.

```
#include <message_parser.hpp>
```

Public Member Functions

- [MessageParser](#) ()
Constructs a [MessageParser](#) and initializes message handlers.
- void [parse_and_print](#) (const std::string &raw_msg)
Parses a raw message and routes it to appropriate handler.

Private Member Functions

- void [handle_auth](#) (const [nlohmann::json](#) &msg)
Handles authentication response messages.
- void [handle_buy](#) (const [nlohmann::json](#) &msg)
Handles buy order response messages.
- void [handle_cancel](#) (const [nlohmann::json](#) &msg)
Handles order cancellation response messages.
- void [handle_subscription](#) (const [nlohmann::json](#) &msg)
Handles subscription response messages.
- void [handle_cancel_all](#) (const [nlohmann::json](#) &msg)
Handles cancel all orders response messages.
- void [handle_positions](#) (const [nlohmann::json](#) &msg)
Handles position information response messages.
- void [handle_orderbook](#) (const [nlohmann::json](#) &msg)
Handles orderbook data response messages.
- void [handle_modify](#) (const [nlohmann::json](#) &msg)
Handles order modification response messages.
- void [handle_unsubscribe](#) (const [nlohmann::json](#) &msg)
Handles unsubscribe response messages.
- void [handle_subscription_response](#) (const [nlohmann::json](#) &msg)
Handles real-time subscription update messages.

Private Attributes

- std::unordered_map< std::string, std::function< void(const [nlohmann::json](#) &)> > [handlers_](#)
Map of message handlers indexed by message type.

3.3.1 Detailed Description

Parser for handling Deribit WebSocket API messages Provides methods to parse and handle different types of responses from the exchange.

3.3.2 Constructor & Destructor Documentation

3.3.2.1 MessageParser()

```
MessageParser::MessageParser ( )
```

Constructs a [MessageParser](#) and initializes message handlers.

Constructs [MessageParser](#) and initializes handler function map.

3.3.3 Member Function Documentation

3.3.3.1 handle_auth()

```
void MessageParser::handle_auth (
    const nlohmann::json & msg ) [private]
```

Handles authentication response messages.

Handles authentication response.

Parameters

<i>msg</i>	The parsed JSON authentication response
------------	---

Processes successful auth and error responses

Parameters

<i>msg</i>	JSON message containing auth result
------------	-------------------------------------

3.3.3.2 handle_buy()

```
void MessageParser::handle_buy (
    const nlohmann::json & msg ) [private]
```

Handles buy order response messages.

Processes buy order response messages.

Parameters

<i>msg</i>	The parsed JSON buy order response
------------	------------------------------------

Extracts and displays order details including ID, amount, price

Parameters

<i>msg</i>	JSON message containing order details
------------	---------------------------------------

3.3.3.3 `handle_cancel()`

```
void MessageParser::handle_cancel (
    const nlohmann::json & msg ) [private]
```

Handles order cancellation response messages.

Processes order cancellation responses.

Parameters

<i>msg</i>	The parsed JSON cancel order response
------------	---------------------------------------

Shows cancellation status and order details

Parameters

<i>msg</i>	JSON message containing cancellation result
------------	---

3.3.3.4 `handle_cancel_all()`

```
void MessageParser::handle_cancel_all (
    const nlohmann::json & msg ) [private]
```

Handles cancel all orders response messages.

Handles bulk order cancellation response.

Parameters

<i>msg</i>	The parsed JSON cancel all response
------------	-------------------------------------

Shows number of orders cancelled

Parameters

<i>msg</i>	JSON message with cancel_all result
------------	-------------------------------------

3.3.3.5 handle_modify()

```
void MessageParser::handle_modify (
    const nlohmann::json & msg ) [private]
```

Handles order modification response messages.

Handles order modification response.

Parameters

<i>msg</i>	The parsed JSON modify order response
------------	---------------------------------------

Shows updated order parameters

Parameters

<i>msg</i>	JSON message containing modified order details
------------	--

3.3.3.6 handle_orderbook()

```
void MessageParser::handle_orderbook (
    const nlohmann::json & msg ) [private]
```

Handles orderbook data response messages.

Processes orderbook data response.

Parameters

<i>msg</i>	The parsed JSON orderbook response
------------	------------------------------------

Displays bids, asks, and market statistics

Parameters

<i>msg</i>	JSON message containing orderbook data
------------	--

3.3.3.7 handle_positions()

```
void MessageParser::handle_positions (
    const nlohmann::json & msg ) [private]
```

Handles position information response messages.

Processes position information response.

Parameters

<i>msg</i>	The parsed JSON positions response
------------	------------------------------------

Shows position size, PNL, and margin details

Parameters

<i>msg</i>	JSON message containing position data
------------	---------------------------------------

3.3.3.8 handle_subscription()

```
void MessageParser::handle_subscription (
    const nlohmann::json & msg ) [private]
```

Handles subscription response messages.

Handles subscription confirmation messages.

Parameters

<i>msg</i>	The parsed JSON subscription response
------------	---------------------------------------

Shows successfully subscribed channels

Parameters

<i>msg</i>	JSON message containing subscription result
------------	---

3.3.3.9 handle_subscription_response()

```
void MessageParser::handle_subscription_response (
    const nlohmann::json & msg ) [private]
```

Handles real-time subscription update messages.

Processes real-time subscription updates.

Parameters

<i>msg</i>	The parsed JSON subscription update
------------	-------------------------------------

Parses and displays orderbook changes and market data

Parameters

<i>msg</i>	JSON message containing subscription update data
------------	--

3.3.3.10 handle_unsubscribe()

```
void MessageParser::handle_unsubscribe (
    const nlohmann::json & msg ) [private]
```

Handles unsubscribe response messages.

Processes unsubscribe response messages.

Parameters

<i>msg</i>	The parsed JSON unsubscribe response
------------	--------------------------------------

Confirms channel unsubscription

Parameters

<i>msg</i>	JSON message containing unsubscribe result
------------	--

3.3.3.11 parse_and_print()

```
void MessageParser::parse_and_print (
    const std::string & raw_msg )
```

Parses a raw message and routes it to appropriate handler.

Parses JSON message and routes to appropriate handler.

Parameters

<i>raw_msg</i>	The raw JSON message string from WebSocket
----------------	--

Extracts function name from message ID and calls corresponding handler

Parameters

<i>raw_msg</i>	Raw JSON string from WebSocket
----------------	--------------------------------

3.3.4 Member Data Documentation

3.3.4.1 handlers_

```
std::unordered_map<std::string, std::function<void(const nlohmann::json&)> > MessageParser↵  
::handlers_ [private]
```

Map of message handlers indexed by message type.

The documentation for this class was generated from the following files:

- include/[message_parser.hpp](#)
- src/[message_parser.cpp](#)

Chapter 4

File Documentation

4.1 build/CMakeCache.txt File Reference

4.2 build/CMakeFiles/3.22.1/CompilerIdC/CMakeCCompilerId.c File Reference

Macros

- `#define __has_include(x) 0`
- `#define COMPILER_ID ""`
- `#define STRINGIFY_HELPER(X) #X`
- `#define STRINGIFY(X) STRINGIFY_HELPER(X)`
- `#define PLATFORM_ID`
- `#define ARCHITECTURE_ID`
- `#define DEC(n)`
- `#define HEX(n)`
- `#define C_VERSION`

Functions

- `int main (int argc, char *argv[])`

Variables

- `char const * info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"`
- `char const * info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"`
- `char const * info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"`
- `const char * info_language_standard_default`
- `const char * info_language_extensions_default`

4.2.1 Macro Definition Documentation

4.2.1.1 __has_include

```
#define __has_include(
    x ) 0
```

4.2.1.2 ARCHITECTURE_ID

```
#define ARCHITECTURE_ID
```

4.2.1.3 C_VERSION

```
#define C_VERSION
```

4.2.1.4 COMPILER_ID

```
#define COMPILER_ID ""
```

4.2.1.5 DEC

```
#define DEC(
    n )
```

Value:

```
('0' + ((n) / 10000000) % 10), \
('0' + ((n) / 1000000) % 10), \
('0' + ((n) / 100000) % 10), \
('0' + ((n) / 10000) % 10), \
('0' + ((n) / 1000) % 10), \
('0' + ((n) / 100) % 10), \
('0' + ((n) / 10) % 10), \
('0' + ((n) % 10))
```

4.2.1.6 HEX

```
#define HEX(
    n )
```

Value:

```
('0' + ((n) >> 28 & 0xF)), \
('0' + ((n) >> 24 & 0xF)), \
('0' + ((n) >> 20 & 0xF)), \
('0' + ((n) >> 16 & 0xF)), \
('0' + ((n) >> 12 & 0xF)), \
('0' + ((n) >> 8 & 0xF)), \
('0' + ((n) >> 4 & 0xF)), \
('0' + ((n) & 0xF))
```

4.2.1.7 PLATFORM_ID

```
#define PLATFORM_ID
```

4.2.1.8 STRINGIFY

```
#define STRINGIFY(  
    X ) STRINGIFY_HELPER(X)
```

4.2.1.9 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(  
    X ) #X
```

4.2.2 Function Documentation

4.2.2.1 main()

```
int main (  
    int argc,  
    char * argv[] )
```

4.2.3 Variable Documentation

4.2.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

4.2.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

4.2.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
= "INFO" ":" "extensions_default["  
  "OFF"  
"]"
```

4.2.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
=  
  "INFO" ":" "standard_default[" C_VERSION "]"
```

4.2.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

4.3 build/CMakeFiles/3.31.4/CompilerIdC/CMakeCCompilerId.c File Reference

Macros

- #define `__has_include(x)` 0
- #define `COMPILER_ID` ""
- #define `STRINGIFY_HELPER(X)` #X
- #define `STRINGIFY(X)` `STRINGIFY_HELPER(X)`
- #define `PLATFORM_ID`
- #define `ARCHITECTURE_ID`
- #define `DEC(n)`
- #define `HEX(n)`
- #define `C_STD_99` 199901L
- #define `C_STD_11` 201112L
- #define `C_STD_17` 201710L
- #define `C_STD_23` 202311L
- #define `C_VERSION`

Functions

- int `main` (int argc, char *argv[])

Variables

- char const * [info_compiler](#) = "INFO" ":" "compiler[" COMPILER_ID "]"
- char const * [info_platform](#) = "INFO" ":" "platform[" PLATFORM_ID "]"
- char const * [info_arch](#) = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
- const char * [info_language_standard_default](#)
- const char * [info_language_extensions_default](#)

4.3.1 Macro Definition Documentation

4.3.1.1 `__has_include`

```
#define __has_include(  
    x ) 0
```

4.3.1.2 `ARCHITECTURE_ID`

```
#define ARCHITECTURE_ID
```

4.3.1.3 `C_STD_11`

```
#define C_STD_11 201112L
```

4.3.1.4 `C_STD_17`

```
#define C_STD_17 201710L
```

4.3.1.5 `C_STD_23`

```
#define C_STD_23 202311L
```

4.3.1.6 C_STD_99

```
#define C_STD_99 199901L
```

4.3.1.7 C_VERSION

```
#define C_VERSION
```

4.3.1.8 COMPILER_ID

```
#define COMPILER_ID ""
```

4.3.1.9 DEC

```
#define DEC(  
    n )
```

Value:

```
('0' + ((n) / 10000000) % 10), \
('0' + ((n) / 1000000) % 10), \
('0' + ((n) / 100000) % 10), \
('0' + ((n) / 10000) % 10), \
('0' + ((n) / 1000) % 10), \
('0' + ((n) / 100) % 10), \
('0' + ((n) / 10) % 10), \
('0' + ((n) % 10))
```

4.3.1.10 HEX

```
#define HEX(  
    n )
```

Value:

```
('0' + ((n) >> 28 & 0xF)), \
('0' + ((n) >> 24 & 0xF)), \
('0' + ((n) >> 20 & 0xF)), \
('0' + ((n) >> 16 & 0xF)), \
('0' + ((n) >> 12 & 0xF)), \
('0' + ((n) >> 8 & 0xF)), \
('0' + ((n) >> 4 & 0xF)), \
('0' + ((n) & 0xF))
```

4.3.1.11 PLATFORM_ID

```
#define PLATFORM_ID
```

4.3.1.12 STRINGIFY

```
#define STRINGIFY(  
    X ) STRINGIFY_HELPER(X)
```

4.3.1.13 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(  
    X ) #X
```

4.3.2 Function Documentation

4.3.2.1 main()

```
int main (  
    int argc,  
    char * argv[] )
```

4.3.3 Variable Documentation

4.3.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

4.3.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

4.3.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
= "INFO" ":" "extensions_default["  
  "OFF"  
"]"
```

4.3.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
=  
  "INFO" ":" "standard_default[" C_VERSION "]"
```

4.3.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

4.4 build/CMakeFiles/3.22.1/CompilerIdCXX/CMakeCXXCompilerId.cpp File Reference

Macros

- `#define __has_include(x) 0`
- `#define COMPILER_ID ""`
- `#define STRINGIFY_HELPER(X) #X`
- `#define STRINGIFY(X) STRINGIFY_HELPER(X)`
- `#define PLATFORM_ID`
- `#define ARCHITECTURE_ID`
- `#define DEC(n)`
- `#define HEX(n)`
- `#define CXX_STD __cplusplus`

Functions

- `int main (int argc, char *argv[ ])`

Variables

- `char const * info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"`
- `char const * info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"`
- `char const * info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"`
- `const char * info_language_standard_default`
- `const char * info_language_extensions_default`

4.4.1 Macro Definition Documentation

4.4.1.1 `__has_include`

```
#define __has_include(  
    x ) 0
```

4.4.1.2 `ARCHITECTURE_ID`

```
#define ARCHITECTURE_ID
```

4.4.1.3 `COMPILER_ID`

```
#define COMPILER_ID ""
```

4.4.1.4 `CXX_STD`

```
#define CXX_STD __cplusplus
```

4.4.1.5 `DEC`

```
#define DEC(  
    n )
```

Value:

```
('0' + ((n) / 10000000) % 10), \  
( '0' + ((n) / 1000000) % 10), \  
( '0' + ((n) / 100000) % 10), \  
( '0' + ((n) / 10000) % 10), \  
( '0' + ((n) / 1000) % 10), \  
( '0' + ((n) / 100) % 10), \  
( '0' + ((n) / 10) % 10), \  
( '0' + ((n) % 10))
```

4.4.1.6 HEX

```
#define HEX(  
    n )
```

Value:

```
('0' + ((n)>>28 & 0xF)), \  
( '0' + ((n)>>24 & 0xF)), \  
( '0' + ((n)>>20 & 0xF)), \  
( '0' + ((n)>>16 & 0xF)), \  
( '0' + ((n)>>12 & 0xF)), \  
( '0' + ((n)>>8  & 0xF)), \  
( '0' + ((n)>>4  & 0xF)), \  
( '0' + ((n)    & 0xF))
```

4.4.1.7 PLATFORM_ID

```
#define PLATFORM_ID
```

4.4.1.8 STRINGIFY

```
#define STRINGIFY(  
    X ) STRINGIFY\_HELPER(X)
```

4.4.1.9 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(  
    X ) #X
```

4.4.2 Function Documentation

4.4.2.1 main()

```
int main (  
    int argc,  
    char * argv[] )
```

4.4.3 Variable Documentation

4.4.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

4.4.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

4.4.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
= "INFO" ":" "extensions_default["
  "OFF"
"]"
```

4.4.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
= "INFO" ":" "standard_default["
  "98"
"]"
```

4.4.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

4.5 build/CMakeFiles/3.31.4/CompilerIdCXX/CMakeCXXCompilerId.cpp File Reference

Macros

- `#define __has_include(x) 0`
- `#define COMPILER_ID ""`
- `#define STRINGIFY_HELPER(X) #X`
- `#define STRINGIFY(X) STRINGIFY_HELPER(X)`
- `#define PLATFORM_ID`
- `#define ARCHITECTURE_ID`
- `#define DEC(n)`
- `#define HEX(n)`
- `#define CXX_STD_98 199711L`
- `#define CXX_STD_11 201103L`
- `#define CXX_STD_14 201402L`
- `#define CXX_STD_17 201703L`
- `#define CXX_STD_20 202002L`
- `#define CXX_STD_23 202302L`
- `#define CXX_STD __cplusplus`

Functions

- int [main](#) (int argc, char *argv[])

Variables

- char const * [info_compiler](#) = "INFO" ":" "compiler[" COMPILER_ID "]"
- char const * [info_platform](#) = "INFO" ":" "platform[" PLATFORM_ID "]"
- char const * [info_arch](#) = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
- const char * [info_language_standard_default](#)
- const char * [info_language_extensions_default](#)

4.5.1 Macro Definition Documentation

4.5.1.1 __has_include

```
#define __has_include(  
    x ) 0
```

4.5.1.2 ARCHITECTURE_ID

```
#define ARCHITECTURE_ID
```

4.5.1.3 COMPILER_ID

```
#define COMPILER_ID ""
```

4.5.1.4 CXX_STD

```
#define CXX_STD __cplusplus
```

4.5.1.5 CXX_STD_11

```
#define CXX_STD_11 201103L
```

4.5.1.6 CXX_STD_14

```
#define CXX_STD_14 201402L
```

4.5.1.7 CXX_STD_17

```
#define CXX_STD_17 201703L
```

4.5.1.8 CXX_STD_20

```
#define CXX_STD_20 202002L
```

4.5.1.9 CXX_STD_23

```
#define CXX_STD_23 202302L
```

4.5.1.10 CXX_STD_98

```
#define CXX_STD_98 199711L
```

4.5.1.11 DEC

```
#define DEC(  
    n )
```

Value:

```
('0' + ((n) / 10000000) % 10), \  
( '0' + ((n) / 1000000) % 10), \  
( '0' + ((n) / 100000) % 10), \  
( '0' + ((n) / 10000) % 10), \  
( '0' + ((n) / 1000) % 10), \  
( '0' + ((n) / 100) % 10), \  
( '0' + ((n) / 10) % 10), \  
( '0' + ((n) % 10))
```

4.5.1.12 HEX

```
#define HEX(  
    n )
```

Value:

```
('0' + ((n)>>28 & 0xF)), \
('0' + ((n)>>24 & 0xF)), \
('0' + ((n)>>20 & 0xF)), \
('0' + ((n)>>16 & 0xF)), \
('0' + ((n)>>12 & 0xF)), \
('0' + ((n)>>8  & 0xF)), \
('0' + ((n)>>4  & 0xF)), \
('0' + ((n)    & 0xF))
```

4.5.1.13 PLATFORM_ID

```
#define PLATFORM_ID
```

4.5.1.14 STRINGIFY

```
#define STRINGIFY(  
    X ) STRINGIFY\_HELPER(X)
```

4.5.1.15 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(  
    X ) #X
```

4.5.2 Function Documentation

4.5.2.1 main()

```
int main (  
    int argc,  
    char * argv[] )
```

4.5.3 Variable Documentation

4.5.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

4.5.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

4.5.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
= "INFO" ":" "extensions_default["  
  "OFF"  
"]"
```

4.5.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
= "INFO" ":" "standard_default["  
  "98"  
"]"
```

4.5.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

4.6 build/CMakeFiles/DeribitClient.dir/link.txt File Reference

Variables

- usr bin c O3 DNDEBUG CMakeFiles DeribitClient dir src deribit_client.cpp o CMakeFiles DeribitClient dir src message_parser.cpp o CMakeFiles DeribitClient dir main.cpp o o DeribitClient usr lib x86_64 linux gnu libboost_system.so usr lib x86_64 linux gnu libboost_thread.so usr lib x86_64 linux gnu libssl.so usr lib x86_64 linux gnu libcrypto.so usr lib x86_64 linux gnu libspdlog.so usr lib x86_64 linux gnu libfmt.so libpthread usr lib x86_64 linux gnu libboost_atomic.so [WI](#)

4.6.1 Variable Documentation

4.6.1.1 WI

```
usr bin c O3 DNDEBUG CMakeFiles DeribitClient dir src deribit_client cpp o CMakeFiles DeribitClient
dir src message_parser cpp o CMakeFiles DeribitClient dir main cpp o o DeribitClient usr lib
x86_64 linux gnu libboost_system so usr lib x86_64 linux gnu libboost_thread so usr lib x86_64
linux gnu libssl so usr lib x86_64 linux gnu libcrypto so usr lib x86_64 linux gnu libspdlog
so usr lib x86_64 linux gnu libfmt so lpthread usr lib x86_64 linux gnu libboost_atomic so Wl
```

4.7 build/CMakeFiles/DeribitClient.dir/main.cpp.o.d File Reference

4.8 build/CMakeFiles/DeribitClient.dir/src/deribit_client.cpp.o.d File Reference

4.9 build/CMakeFiles/DeribitClient.dir/src/message_parser.cpp.o.d File Reference

4.10 build/CMakeFiles/TargetDirectories.txt File Reference

4.11 build_and_run.sh File Reference

4.12 CMakeLists.txt File Reference

Functions

- `cmake_minimum_required` (VERSION 3.16) `project(DeribitClient)` `set(CMAKE_CXX_STANDARD 17)` `set(CMAKE_CXX_STANDARD_REQUIRED ON)` `if(NOT CMAKE_BUILD_TYPE)` `set(CMAKE_BUILD_TYPE Release)` `endif()` `set(SOURCES src/deribit_client.cpp src/message_parser.cpp main.cpp)` `include_directories(include)` `find_package(Boost REQUIRED COMPONENTS system thread)` `find_package(OpenSSL REQUIRED)` `find_package(spdlog REQUIRED)` `find_package(fmt REQUIRED)` `add_executable($`

4.12.1 Function Documentation

4.12.1.1 `cmake_minimum_required()`

```
cmake_minimum_required (
    VERSION 3. 16 )
```


4.13 include/deribit_client.hpp File Reference

```
#include <boost/beast/core.hpp>
#include <boost/beast/ssl.hpp>
#include <boost/beast/websocket.hpp>
#include <boost/asio/ip/tcp.hpp>
#include <boost/asio/ssl/context.hpp>
#include <boost/asio/io_context.hpp>
#include <nlohmann/json.hpp>
#include <mutex>
#include <string>
#include <thread>
#include <optional>
#include "message_parser.hpp"
#include <unordered_set>
#include <queue>
#include <spdlog/spdlog.h>
#include <spdlog/sinks/rotating_file_sink.h>
#include <spdlog/sinks/stdout_color_sinks.h>
#include <boost/asio/strand.hpp>
#include <cstdlib>
#include <memory>
#include <set>
```

Include dependency graph for deribit_client.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [DeribitClient](#)
Client class for interacting with the Deribit cryptocurrency exchange API.
- struct [DeribitClient::ClientSession](#)

Typedefs

- using [tcp](#) = boost::asio::ip::tcp
- using [json](#) = nlohmann::json

4.13.1 Typedef Documentation

4.13.1.1 json

```
using json = nlohmann::json
```

4.13.1.2 tcp

```
using tcp = boost::asio::ip::tcp
```

4.14 include/message_parser.hpp File Reference

```
#include <string>
#include <unordered_map>
#include <functional>
#include <nlohmann/json.hpp>
#include <fmt/format.h>
#include <fmt/core.h>
```

Include dependency graph for message_parser.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [MessageParser](#)

Parser for handling Deribit WebSocket API messages Provides methods to parse and handle different types of responses from the exchange.

4.15 include/prompts.hpp File Reference

```
#include <string>
#include <iostream>
#include "deribit_client.hpp"
```

Include dependency graph for prompts.hpp: This graph shows which files directly or indirectly include this file:

Functions

- void [get_place_order_prompt](#) ([DeribitClient](#) &client)
Handles user input for placing a new order.
- void [get_modify_order_prompt](#) ([DeribitClient](#) &client)
Handles user input for modifying an existing order.
- void [get_orderbook_prompt](#) ([DeribitClient](#) &client)
Handles user input for fetching orderbook data.
- void [get_positions_prompt](#) ([DeribitClient](#) &client)
Handles user input for fetching position information.
- void [get_unsubscribe_prompt](#) ([DeribitClient](#) &client)
Handles user input for unsubscribing from market data.

4.15.1 Function Documentation

4.15.1.1 [get_modify_order_prompt\(\)](#)

```
void get_modify_order_prompt (
    DeribitClient & client )
```

Handles user input for modifying an existing order.

Parameters

<i>client</i>	Reference to the DeribitClient instance
---------------	---

Prompts user for:

- Order ID (required)
- New amount and price
- Post-only setting
- Time in force
- Label

4.15.1.2 `get_orderbook_prompt()`

```
void get_orderbook_prompt (  
    DeribitClient & client )
```

Handles user input for fetching orderbook data.

Parameters

<i>client</i>	Reference to the DeribitClient instance
---------------	---

Prompts user for:

- Symbol (required)
- Orderbook depth (optional, defaults to 10)

4.15.1.3 `get_place_order_prompt()`

```
void get_place_order_prompt (  
    DeribitClient & client )
```

Handles user input for placing a new order.

Parameters

<i>client</i>	Reference to the DeribitClient instance
---------------	---

Prompts user for order details including:

- Symbol (required)
- Amount and price
- Order type (limit/market)
- Time in force
- Label
- Trigger settings
- Post-only and reduce-only flags

4.15.1.4 `get_positions_prompt()`

```
void get_positions_prompt (
    DeribitClient & client )
```

Handles user input for fetching position information.

Parameters

<i>client</i>	Reference to the DeribitClient instance
---------------	---

Prompts user for:

- Currency (required, e.g., "BTC")
- Position kind (optional, defaults to "any")

4.15.1.5 `get_unsubscribe_prompt()`

```
void get_unsubscribe_prompt (
    DeribitClient & client )
```

Handles user input for unsubscribing from market data.

Parameters

<i>client</i>	Reference to the DeribitClient instance
---------------	---

Prompts user for:

- Symbol to unsubscribe from (required)

4.16 include/websocket_server.hpp File Reference

4.17 main.cpp File Reference

```
#include "deribit_client.hpp"
#include <iostream>
#include <string>
#include <prompts.hpp>
Include dependency graph for main.cpp:
```

Functions

- void `print_menu` ()
Displays the main menu options to the user.
- int `main` ()
Main entry point for the Deribit WebSocket client.

4.17.1 Function Documentation

4.17.1.1 `main()`

```
int main ( )
```

Main entry point for the Deribit WebSocket client.

Initializes the client, handles the main menu loop, and provides error handling for the application

Returns

0 on successful execution, non-zero on error

4.17.1.2 `print_menu()`

```
void print_menu ( )
```

Displays the main menu options to the user.

Shows available operations including:

- Authentication
- List subscriptions
- Order operations (place, modify, cancel)
- Market data operations (orderbook, positions)

4.18 src/deribit_client.cpp File Reference

Implementation of the [DeribitClient](#) class for interacting with Deribit exchange.

```
#include "deribit_client.hpp"
#include <boost/asio/connect.hpp>
#include <iostream>
Include dependency graph for deribit_client.cpp:
```

4.18.1 Detailed Description

Implementation of the [DeribitClient](#) class for interacting with Deribit exchange.

4.19 src/message_parser.cpp File Reference

Implementation of the [MessageParser](#) class for handling Deribit API messages.

```
#include "message_parser.hpp"
#include <iostream>
#include <iomanip>
Include dependency graph for message_parser.cpp:
```

Typedefs

- using [json](#) = nlohmann::json

4.19.1 Detailed Description

Implementation of the [MessageParser](#) class for handling Deribit API messages.

4.19.2 Typedef Documentation

4.19.2.1 json

```
using json = nlohmann::json
```